

Ahtesham Alvi

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EDUCATION

University of Maryland, College Park | College Park, MD

Expected May 2027

Bachelor of Science, Computer Science — *Honors College*

Bachelor of Science, Finance — *Robert H. Smith School of Business*

Minors: Data Science; Robotics and Autonomous Systems

Relevant Coursework: Machine Learning (CMSC422), Algorithm Design & Analysis (CMSC451), Computer Networks (CMSC417), Data Science (CMSC320), Advanced Data Structures (CMSC420), Software Engineering (CMSC435), Honors Research Seminar (CMSC396H), Independent Undergraduate Research (CMSC499A), Robotics Programming (ENAE450), Introduction to Robotics (ENME480), Robotics Project Laboratory (ENEE467), Differential Equations (MATH246)

AWARDS & HONORS

University Honors College, University of Maryland

2023 – Present

Semester Academic Honors, University of Maryland | Fall 2023, Spring 2024, Fall 2024

First-Year Innovation & Research Experience (FIRE) — program completion, Bio-Inspired Robotics stream

2024

RESEARCH EXPERIENCE & PUBLICATIONS

De Novo Protein Binder Design for IBD Therapeutics (IL-23)

March 2026 – Present

Independent Undergraduate Research (CMSC499A) | Advisor: Prof. Fardina Alam | University of Maryland

- Contributed to a two-student team designing de novo miniprotein binders against the IL-23R-binding face of the IL-23 p19 subunit (PDB 3DUH), a target not previously addressed by de novo design — prior miniprotein work targets the IL-23 receptor rather than the cytokine itself.
- Built the project's computational scaffold and co-designed its first pipeline arm** — a 9-stage RFdiffusion → ProteinMPNN → ColabFold workflow on SLURM HPC — and ran and analyzed the resulting screen, which produced no reliable binder across 972 designs (best ipSAE 0.30, none above the 0.60 reliable-interface threshold).
- Diagnosed the pessimistic single-sequence ColabFold signal from that screen as a *scoring artifact* rather than evidence of unfoldable designs: a ~56-point binder-pLDDT under-rating relative to AlphaFold initial-guess on identical backbones.
- Second pipeline arm, led by the primary author, applied interface-optimized hallucination (BindCraft) under the same scoring path, yielding 42 of 50 designs above the 0.60 threshold and five all- α -helical leads of 72–104 residues with Rosetta interface energies of –57 to –87 REU.
- First-authored** a follow-on study superposing all 50 accepted designs into the experimental IL-23:IL-23R assembly: all 50 overlap the IL-23R-bound region, but 45 also clash with the obligate p40 subunit (0 to >3,500 atom pairs) — a constraint none of nine interface-confidence metrics tracks, corroborated by a sequence-only Boltz-2 analysis ($\rho = -0.60$).

[1] R. Paladugu, F. Alam, and A. Alvi. “De Novo Design of Miniprotein Candidate Neutralizers Targeting the IL-23 Cytokine Subunit p19: A Rigorously Scored Pipeline Comparison.” *Under review, IEEE International Conference on Bioinformatics and Biomedicine (BIBM) 2026*.

- [2] A. Alvi and F. Alam. “What Interface Confidence Cannot See: Assembly-Level Steric Validation of De Novo IL-23 p19 Binder Designs.” *Manuscript in preparation (first author)*.

Temporal Robustness of Social Bot Detection | [GitHub](#)

April 2026 – May 2026

Honors Research Seminar (CMSC396H) | University of Maryland

- Engineered account, behavioral, graph (PageRank, reciprocity, ego density), and node2vec features over four Twitter/X bot benchmarks — Cresci-2015, Cresci-2017, TwiBot-20, and TwiBot-22 — spanning 2015–2022 (up to 1M users and 88M tweets), normalized into a shared schema in DuckDB, NetworkX, and pandas.
 - Trained six supervised classifiers (logistic regression, ridge, lasso, linear SVM, random forest, XGBoost) across all 16 train/test dataset pairs under a per-pair feature contract, yielding a 4×4 cross-era transfer matrix per classifier.
 - Found that every classifier attains near-perfect in-distribution AUC (up to 0.9955) yet falls below 0.60 mean AUC across datasets, and that model *family* governs transfer behavior more than algorithm choice: the four linear models behave near-identically while the two tree-based models follow a distinct pattern.
 - Attributed the generalization gap to dataset construction, labeling, and class balance alongside bot evolution, since models trained on newer data sometimes transferred worse to older data — indicating single-benchmark accuracy overstates real-world reliability.
- [3] A. Alvi, M. Morton, D. Scott, Y. Senthikumar, and T. Tang. “Evaluating the Temporal Robustness of Twitter Bot Detection Models.” *Course research paper, CMSC396H, University of Maryland, 2026*.

SeaTerp — Bioinspired Turtle Robot for Chesapeake Bay Dead-Zone August 2024 – December 2024 Monitoring | [GitHub](#)

FIRE Bio-Inspired Robotics (FIRE298) | Advisor: Dr. Lena Johnson | University of Maryland

- Co-developed a tethered, sea-turtle-inspired underwater robot to survey dissolved-oxygen levels in the Chesapeake Bay dead zone, where nutrient runoff drives algal blooms that strip oxygen from the water column.
 - Built the locomotion and sensing stack on an Arduino Uno driving six DC motors through two Adafruit FeatherWing controllers — crankshaft-actuated flippers, twin rear propellers, and a two-motor buoyancy bladder — instrumented with a dissolved-solids sensor and an onboard camera.
 - Iterated the hull across successive CAD and 3D-printed revisions, moving from a rounded profile to a flatter ballast form for hydrodynamic stability.
 - Validated subsystems over 20 bench trials spanning buoyancy control, dissolved-solids sensing at three water qualities, crankshaft actuation, and differential propulsion; the robot achieved 360° maneuvering and sampled water quality, with an underwater trial ending early on hull ingress.
- [4] A. Srivatsa, T. Zhang, A. Alvi, G. Khawaja, and L. Johnson. “Exploring the Use of Bioinspired Robots in Treating the Chesapeake Bay Dead Zone.” *Course conference paper, FIRE298, University of Maryland, 2024*.

RoboRaptor — Articulated-Wing Gliding Robot | [GitHub](#)

May 2024 – July 2024

FIRE Research Internship (FIRE199) | Advisor: Dr. Lena Johnson | University of Maryland

- Led a three-person team building a bird-inspired tethered gliding robot with articulating flapping wings, instrumented to characterize wind and atmospheric effects on flapping and gliding flight.
- Advanced the design across four prototypes benchmarked against house sparrow, peregrine falcon, and golden eagle wing geometry, raising the model’s aspect ratio from 2.92 to 6.24 and replacing wooden dowel spars with carbon fiber and aluminum after in-flight structural failures.
- Wrote Arduino firmware for servo wing-torque control and SD-card flight telemetry, integrating an onboard wind sensor on a 3D-printed electronics plate.
- Achieved sustained flight in winds above 7 mph with in-flight wing actuation and directional control.

CONFERENCE PRESENTATIONS

“Sea Terp.” Poster presentation, *The FIRE Summit*, Colony Ballroom and Charles Carroll Room, Stamp Student Union, University of Maryland. **December 2024**
A. Alvi, G. Khawaja, A. Srivatsa, and T. Zhang — first author

“RoboRaptor: Articulated Wing Glider.” Poster presentation, *Summer Undergraduate Research Conference (SURC)*, Grand Ballroom, Stamp Student Union, University of Maryland. **July 19, 2024**
A. Alvi, P. Sethy, S. Jayanthi, and L. Johnson — first author

TEACHING EXPERIENCE

Code Ninjas | Code Sensei | Yardley, PA **June 2022 – December 2022**

- Taught JavaScript fundamentals to over 30 students aged 7–14 through project-based instruction built around games and coding tasks.
- Mentored students through independent projects, reviewing their code and giving iterative feedback on structure and problem decomposition.

PROFESSIONAL EXPERIENCE

fusionSpan | Software Developer Intern | Rockville, MD **June 2026 – August 2026**

Salesforce platform engineering for national trade associations and nonprofits

- Diagnosed and repaired more than 50 client-reported and backlog defects over an 11-week term, working tickets under per-ticket senior-developer code review.
- Wrote and modernized dozens of Apex test classes across eight association and nonprofit client organizations, realigning legacy test suites with the current state of each codebase.
- Built Jenkins test environments and deployed test-suite updates and validation-rule changes into multiple production orgs.
- Designed a five-skill Claude agent toolchain spanning the ticket lifecycle: persistent cross-session ticket notes, PMD static-analysis triage and repair for Apex, Confluence documentation generated from git diffs rather than conversation history, and a categorized code library with an automated consistency audit.

Urban Food Alliance | Software Development Intern | Remote **July 2025 – November 2025**

- Built a financial-literacy chatbot in Python with an NLP intent and response pipeline, broadening access to financial guidance for underserved communities.
- Improved query understanding and response accuracy through iterative evaluation of dialogue quality, working in a Scrum cadence with code review and pre-release testing.

SELECTED TECHNICAL PROJECTS

Nenworld — Data-Driven Rules Engine | [GitHub](#) **July 2026 – Present**

- Architected a pure, side-effect-free rules kernel in TypeScript — 120 files, roughly 38,000 lines, zero runtime dependencies — resolving authored data into derived state across ten content domains, with every public entry point returning a serializable trace explaining how each value was computed. Maintained 596 tests across 27 suites under a strict type-check gate.

Runemaker — Deterministic Procedural 3D Geometry | [GitHub](#) **April 2026**

- Built a deterministic pipeline mapping a word through 125 language translations into a smoothed vector field, tracing streamlines and sweeping them into a tube mesh rendered by a custom Vulkan/GLSL viewer in C++.

QuantumGuard — Quantum Key Distribution Demonstration | [GitHub](#) | [Live](#) **April 2026**

- Implemented BB84 in Qiskit Aer over 256 qubits with AES-128-GCM key derivation and an 11% QBER rejection threshold, deployed as a multi-device application staging a three-party man-in-the-middle attack.

Exoplanet Habitability Classification | [GitHub](#)

January 2026 – June 2026

- Consolidated the NASA Exoplanet Archive, Habitable Exoplanets Catalog, Gaia DR3, and SIMBAD into one dataset, training a semi-supervised three-class model on ~5,600 labeled planets to classify ~600 unlabeled confirmed exoplanets.

Robotic Arm Manipulator Control | [GitHub](#)

August 2025 – December 2025

- Developed a ROS2 manipulation pipeline integrating ArUco perception, homography-based pixel-to-world calibration, inverse kinematics, and autonomous pick-and-place, validated on physical hardware.

Mini C Compiler | [GitHub](#)

May 2025

- Implemented a compiler front end in OCaml and Rust with Hindley–Milner-style type inference (constraint generation, unification, substitution) and multi-pass AST optimization including constant folding, algebraic simplification, and dead-branch elimination.

Additional projects: AST-based Unix shell in C (December 2024) • quantum computing experiments in Qiskit including CHSH inequality tests on IBM Quantum hardware (February–April 2026) • statistical and ML analysis of socioeconomic predictors of college GPA (December 2025)

SERVICE & LEADERSHIP

UMD Student Government Finance Committee | *Budget Reviewer*

September 2024 – Present

- Administer a \$2.7M annual budget across more than 200 campus organizations, reviewing funding requests, adjudicating appeals, and voting on funding legislation.

UMD College of Computer, Mathematical & Natural Sciences | *Office Assistant*

January 2024 – Present

- Archived and digitized over 4,000 documents and audited payroll and tax records in Workday and Excel to IRS compliance standards.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C/C++, Rust, OCaml, SQL, SOQL, Apex, Bash, GLSL, $\LaTeX$

Computational Biology: RFdiffusion, ProteinMPNN, BindCraft, AlphaFold2-Multimer, ColabFold, Boltz-2, Rosetta, ipSAE, PyMOL, BioPython, SLURM/HPC

Machine Learning & Data: PyTorch, scikit-learn, XGBoost, imbalanced-learn, pandas, NumPy, SciPy, NetworkX, node2vec, DuckDB, astroquery

Robotics & Quantum: ROS2 (Humble), Gazebo, OpenCV, Arduino, Qiskit, IBM Quantum hardware

Systems & Web: Linux, Docker, Jenkins, PMD, Git, Vitest, pytest, Vulkan, React, Next.js, FastAPI, WebSockets

REFERENCES

Available upon request.